Exam 3 — Solutions

Synthesis and transfer

Answer key and worked explanations. Source aliases and convention warnings are listed in `COVERAGE.md` and `SOURCE_NOTES.md` in the study pack.

Quick key

Question	Answer	Question	Answer
01	C	26	D
02	C	27	A
03	B	28	D
04	B	29	A
05	A	30	B
06	B	31	260.00
07	D	32	0.26
08	A	33	6.00
09	C	34	0.25
10	D	35	9
11	B	36	10
12	D	37	3
13	A	38	5
14	B	39	4.38
15	B	40	14.62
16	C	41	105
17	A	42	2.33
18	D	43	2.00
19	B	44	0.50
20	A	45	0.57
21	A	46	0.45
22	C	47	4448
23	C	48	275
24	C	49	0.36
25	D	50	0.31

Worked solutions

Q01 — C

C. $n_w = R^\top n_c$; translation does not act on a direction vector.
Normals between rigid coordinate frames transform by rotation only. R maps world directions to camera directions; its transpose maps back. A general non-rigid affine transform would require a different normal rule, which is not asked here.
Topic: camera. Sources: HC1b pp.35–45; HC2a surface normals; prerequisite linear algebra.

Q02 — C

C. High spatial frequencies can otherwise alias into false lower-frequency patterns.
Subsampling reduces the representable frequency range. Filtering first suppresses frequencies that the coarse grid cannot represent; smoothing after aliasing does not undo it.
Topic: filtering. Sources: HC2b pp.93–105; HC5b pp.38–39; Szeliski §3.4.

Q03 — B

B. No; multiple albedo/shading combinations can produce the same image.
For positive scale c, $A' = cA$ and $S' = \frac{S}{c}$ preserve I, and further spatial ambiguities exist. Priors or extra observations are required.
Topic: reflectance. Sources: HC2a pp.4–6,21–23,47–48; Lab1.

Q04 — B

B. The model assumes a linear relation between measured intensity and radiance; the response must be linearized or modelled.
If recorded value is a nonlinear function of radiance, the original linear dot-product equations no longer describe the observations. Camera-response calibration or suitable raw data is needed.
Topic: photometric. Sources: HC2a pp.19–20,61–62; Lab1.

Q05 — A

A. The interaction of source spectrum, surface spectral reflectance and that channel's spectral sensitivity.
A channel integrates wavelength-dependent contributions. Material colour alone does not fix measured RGB; illumination and sensor response matter.
Topic: colour. Sources: HC1b pp.67–71,80–86; Lab1.

Q06 — B

B. An additive constant in height.
Derivatives determine relative height, not the reference height. Fixing a height such as $z(0,0) = 0$ removes this additive ambiguity on a connected region.
Topic: photometric. Sources: HC2a pp.41–43; Lab1; Forsyth–Ponce §2.2.

Q07 — D

D. Interior outputs remain constant, but outputs near boundaries can be darker.
Interior samples see a full constant patch and weights sum to 1. At a zero-padded boundary some weighted values are 0. Padding is part of the filter's actual behaviour.
Topic: filtering. Sources: HC2b pp.47–87; Lab2.

Q08 — A

A. I_t can fill sufficiently small gaps/holes relative to the element's shape, but need not fill a large cavity.
Closing is scale- and shape-dependent. I_t is dilation followed by erosion, not general topological hole filling.
Topic: morphology. Sources: HC2b pp.39–45; S2.

Q09 — C

C. Its outermost pixel border is removed.
For a border pixel, part of the tested support lies outside the image and encounters 0. Interior pixels whose entire support is inside survive.
Topic: morphology. Sources: HC2b pp.27–38; S2.

Q10 — D

D. Each coefficient is multiplied by a frequency-dependent phase factor of unit magnitude; its magnitude is unchanged.
The translated spectrum is the original spectrum times a unit-magnitude complex exponential. This is exact for a circular shift; cropping/zero-filled shifts can also change magnitude through boundaries. A particular coefficient can have an unchanged phase when its factor is 1.
Topic: fourier. Sources: HC2b pp.97–104; Szeliski §3.4.

Q11 — B

B. They are complex conjugates, with indices interpreted modulo the image size.
Real-valued signals have Hermitian symmetry $F_{-k} = \overline{F_k}$. Thus magnitudes are symmetric and phases have opposite signs modulo a full turn, where defined.
Topic: fourier. Sources: HC2b pp.99–104; Szeliski §3.4.

Q12 — D

D. Reflectance or illumination changes can also cause an intensity edge.
Image derivatives measure appearance changes. Material boundaries, shadows, highlights and orientation changes can create edges without a depth break.
Topic: features. Sources: HC3a; S3 introduction; HC2a.

Q13 — A

A. To locate distinctive structures at their characteristic scales; DoG approximates a scale-normalized LoG response.
Scale selection allows corresponding structures to be detected after resizing. DoG is an efficient approximation, not a guarantee of unlimited viewpoint invariance.
Topic: matching. Sources: HC4a pp.36–42; S4; Szeliski §7.1.

Q14 — B

B. Gradients remove the constant offset b, and descriptor normalization reduces the common gain a.
The derivative of $aI + b$ is a times the derivative of I. Orientation and normalized magnitudes are relatively stable, provided clipping, quantization and nonlinear response do not disrupt the assumption.
Topic: matching. Sources: HC4a pp.41–44; Szeliski §7.1.2 pp.434–436.

Q15 — B

B. I_t is geometrically degenerate and does not uniquely constrain a general 2D affine transform.
All-inlier probability alone is not enough. A minimal sample must also have the geometry required to determine the model. For affine transforms, the $[x, y, 1]$ design rows must have rank 3.
Topic: fitting. Sources: HC4a pp.17–22,45–55; Lab4.

Q16 — C

C. The larger threshold can admit poor matches and outliers as inliers.
A threshold controls what residual is accepted. Compare fit quality using meaningful units and tolerances, rather than maximizing a count under arbitrarily relaxed criteria.
Topic: fitting. Sources: HC4a pp.46–48; HC3a pp.99–109.

Q17 — A

A. The appearance-constancy assumption for a physical surface point is violated; apparent image motion need not equal physical surface motion.
Optical flow describes apparent image change and relies on assumptions. Lighting changes can mimic motion even with stationary geometry. A moving shadow pattern may itself have a consistent image displacement; that is not the underlying surface's displacement.
Topic: flow. Sources: HC3b pp.8–47,57–58; Lab3.

Q18 — D

D. The object can disappear or become indistinct at coarse levels, so its motion is not initialized correctly.
Pyramids reduce displacement but also reduce detail. Coarse estimation can miss thin or small structures, and later refinement may remain in the wrong solution basin.
Topic: flow. Sources: HC3b pp.52–58.

Q19 — B

B. Incorrect predictions can contaminate the model and cause tracking drift.
Self-labelled updates can learn the occluder or background. Controlled updates and re-detection can help, but no method is universally immune.
Topic: tracking. Sources: HC3b pp.89–106; Lab3.

Q20 — A

A. Depth-dependent parallax; one global homography need not align the nonplanar scene.
Translation makes image displacement depend on depth. Blending can hide seams but cannot generally correct incompatible geometric alignments or double images.
Topic: geometry. Sources: HC4a pp.33–35,60–64; S4; Lab4.

Q21 — A

A. TH
Matrix composition applies the rightmost transformation first: source $\xrightarrow{H}$ reference $\xrightarrow{T}$ canvas. Transform direction and multiplication order must be explicit.
Topic: geometry. Sources: HC4a pp.19,24–35; Lab4.

Q22 — C

C. Unweighted 3-nearest-neighbour majority voting predicts class 0, although the closest example has class 1.
Squared distances are $\begin{bmatrix}0.02 & 0.82 & 0.82 & 16.82\end{bmatrix}$. The three nearest labels are $\begin{bmatrix}1 & 0 & 0\end{bmatrix}$, in an irrelevant tie order for the two zero labels. Their sum is 1, so the comparison is false and prediction $= 0$. k-NN uses labelled examples; it does not update cluster centres. With $k = 1$ the prediction would instead be 1.
Topic: classification. Sources: HC4b pp.37–53; Lab0.

Q23 — C

C. Spatial terms encourage location coherence, and α sets their relative importance against colour differences.
K-means uses the metric in the supplied feature space. Scaling positions changes distances and can trade colour similarity against spatial compactness; it does not guarantee semantics or connectivity.
Topic: clustering. Sources: HC5a pp.91–118.

Q24 — C

C. I_t permits margin violations while balancing fit against regularization.
A hard-margin separator is infeasible for nonseparable labels. Soft margin penalizes violations and controls model complexity; it does not know which labels are wrong.
Topic: classification. Sources: HC4b pp.69–78.

Q25 — D

D. The probabilities are well calibrated to actual correctness frequencies.
Softmax produces a normalized score distribution, but calibration is a separate empirical property. A high score alone is not proof that its stated frequency matches reality.
Topic: metrics. Sources: HC5a p.5; HC5b p.78; prerequisite probability.

Q26 — D

D. Prevent unintended accumulation of earlier batches' gradients before this batch's update.
PyTorch gradient buffers accumulate unless cleared. Clearing gradients does not reset parameter values or disable gradient recording. Deliberate gradient accumulation is possible, but would need a different specified training schedule.
Topic: optimization. Sources: Lab5 tutorial and training code.

Q27 — A

A. A compatible shortcut, such as a stride-2 projection, so the tensors have matching shapes.
Elementwise residual addition requires compatible dimensions. Identity shortcuts work only when the required shape matches; otherwise use an appropriate projection/downsampling path.
Topic: architectures. Sources: HC6a pp.99–111.

Q28 — D

D. IoU is unchanged, while an unnormalized squared coordinate-error loss generally scales by the square of that factor.
Intersection and union areas both scale by the square factor, so their ratio stays fixed. Squared coordinate errors also scale by the square factor, without cancellation.
Topic: detection. Sources: S5 Exercise 4; YOLOv4 §2.2.

Q29 — A

A. The epipoles are at infinity in the horizontal baseline direction.
The other camera centre projects to a point at infinity when the baseline is parallel to the image plane. Corresponding points lie on horizontal scanlines after rectification.
Topic: stereo. Sources: HC6b pp.50–90.

Q30 — B

B. SfM estimates camera poses and sparse structure; bundle adjustment refines poses and points jointly; MVS can densify using estimated cameras.
SfM recovers geometry from correspondences and motion. Ordinary monocular reconstruction retains a global scale ambiguity without a metric reference. MVS and bundle adjustment have different roles.
Topic: stereo. Sources: HC6b pp.108–149.

Q31 — 260.00

Submit 260.00. Unrounded reference: `260.0`.
After crop and resize: $f'_y = 300$ and $c'_y = \frac{240-20}{2} = 110$. $v' = 300 \cdot \frac{1}{2} + 110 = 260$. Extrinsics do not change.
Topic: camera. Sources: HC1b pp.24–45; S1.

Q32 — 0.26

Submit 0.26. Unrounded reference: `0.2625`.
The per-pixel sums are

$$\begin{bmatrix} 10 & 10 \\ 0 & 4 \end{bmatrix},$$

with the zero denominator replaced by 1. Normalized red values are

$$\begin{bmatrix} 0.6 & 0.2 \\ 0 & 0.25 \end{bmatrix}.$$

Their mean is $\frac{1.05}{4} = .2625$; submit .26. Normalizing a spatially averaged RGB vector would be a different calculation.
Topic: colour. Sources: Lab1 Section 1; Lab0 array axes.

Q33 — 6.00

Submit 6.00. Unrounded reference: `6.0`.
Path A: $p = 0$ along $y = 0$, then $q = 4$ along $x = 2$, giving $z_A = 12$. Path B: $q = 0$ along $x = 0$, then $p = 3$ along $y = 3$, giving $z_B = 6$. Difference $= 6$. The field is not integrable because $\frac{\partial p}{\partial y} = 1$ differs from $\frac{\partial q}{\partial x} = 2$.
Topic: photometric. Sources: HC2a pp.41–43; Lab1.

Q34 — 0.25

Submit 0.25. Unrounded reference: `0.25`.
$n \cdot l = -\frac{1}{3} \cdot .6 + \frac{2}{3} \cdot .8 = \frac{1}{3}$. I_t is positive, so $I = \frac{.75}{3} = .25$.
Topic: reflectance. Sources: HC2a pp.12–20; S1; Lab1.

Q35 — 9

Submit 9. Unrounded reference: `9`.
Erosion leaves only the central pixel (4,4); the isolated corner pixel disappears. Dilation restores the central 3 by 3 block, giving 9 pixels. I_t does not restore the isolated noise.
Topic: morphology. Sources: HC2b pp.39–42; S2.

Q36 — 10

Submit 10. Unrounded reference: `10`.
$a + b = \begin{bmatrix} 0 & 10 & 10 \end{bmatrix}$, whose median is 10. Both original medians are 0. The difference is 10, a concrete failure of additivity proving that median filtering is nonlinear.
Topic: filtering. Sources: HC2b pp.47,89–91; Lab2.

Q37 — 3

Submit 3. Unrounded reference: `3`.
$c_3 = 1h_3 + 0h_2 + 2h_1 + 0h_0 = 1 + 2 = 3$. Wrap-around is part of the defined operation.
Topic: fourier. Sources: HC2b pp.101–103; Szeliski §3.4.

Q38 — 5

Submit 5. Unrounded reference: `5`.
The left group has one strong 10 and four connected weak 6s, so all 5 survive. The three 5s on the right have no connection to a strong seed and are discarded.
Topic: features. Sources: HC3a pp.58–65; S2.

Q39 — 4.38

Submit 4.38. Unrounded reference: `4.381966011250105`.

$$M = \begin{bmatrix} 6 & 1 \\ 1 & 5 \end{bmatrix}.$$

$\lambda_{\min} = \frac{11-\sqrt{5}}{2} = 4.381966$. Submit 4.38. The score tests the least constrained direction.
Topic: features. Sources: S3; Lab3 Sections 2–3.

Q40 — 14.62

Submit 14.62. Unrounded reference: `14.615915982243466`.
Scale coarse displacement by 8: $\begin{bmatrix} 14 & -4 \end{bmatrix}$. Add the residual: $\begin{bmatrix} 14.25 & -3.25 \end{bmatrix}$. Magnitude $= \sqrt{14.25^2 + 3.25^2} \approx 14.62$ pixels. Resizing the array alone is not enough.
Topic: flow. Sources: HC3b pp.48–54; Lab3.

Q41 — 105

Submit 105. Unrounded reference: `105`.
$w^3 = .064$. $\frac{\log(.001)}{\log(.936)} = 104.4417$. Ceiling gives 105. A homography would require a different sample size and a larger budget.
Topic: fitting. Sources: HC3a pp.104–105; Lab4.

Q42 — 2.33

Submit 2.33. Unrounded reference: `2.3333333333333335`.
The first pair gives $c = 1$, $f = 2$. The next pairs imply $a = 3g + 2$, $d = 2g$, $b = h$, $e = 4h + 2$. The last pair gives $h = 0$, $g = 1$. Thus

$$H = \begin{bmatrix} 5 & 0 & 1 \\ 2 & 2 & 2 \\ 1 & 0 & 1 \end{bmatrix}.$$

At (.5,.5), $x' = \frac{3.5}{1.5} = \frac{7}{3}$; submit 2.33. Averaging target corner positions would not implement a projective map.
Topic: geometry. Sources: HC4a pp.47–55; S4.

Q43 — 2.00

Submit 2.00. Unrounded reference: `2.0`.
Ordinary ratios are $\begin{bmatrix}2/3 & 3/4 & 1/2 & 5/6 & 1/2\end{bmatrix}$. Accepted nearest distances are $\begin{bmatrix}2 & 1 & 0.5\end{bmatrix}$, whose maximum is 2. Comparing $\frac{D_1}{D_2}$ directly with .7 would use the wrong threshold.
Topic: matching. Sources: HC4a pp.43–44; S4; Lab4.

Q44 — 0.50

Submit 0.50. Unrounded reference: `0.5`.
Assignments produce counts $\begin{bmatrix}1 & 2 & 3\end{bmatrix}$. After normalization by 6, the final bin is $\frac{3}{6} = .5$. The fixed vocabulary gives the same histogram length for every image.
Topic: classification. Sources: HC4b pp.23–28; S4.

Q45 — 0.57

Submit 0.57. Unrounded reference: `0.5714285714285714`.
The first four predictions are positive. TP = 2, FP = 2, FN = 1. $F_1 = \frac{4}{4+2+1} = \frac{4}{7} = .5714286$; submit .57. The score exactly at .6 is positive because the rule is inclusive.
Topic: metrics. Sources: HC4b pp.80–86; S4.

Q46 — 0.45

Submit 0.45. Unrounded reference: `0.45`.
Cumulative precisions are $\begin{bmatrix}0 & 1/2 & 1/3 & 2/4 & 3/5\end{bmatrix}$, recalls $\begin{bmatrix}0 & 1/4 & 1/4 & 2/4 & 3/4\end{bmatrix}$. The future-maximum precision at each TP is .6. Three increments of .25 contribute $.25 \cdot .6$ each, so AP = .45. The missing final quarter of recall contributes zero; raw uninterpolated AP would instead be .40.
Topic: metrics. Sources: HC5a pp.68–78; S4.

Q47 — 4448

Submit 4448. Unrounded reference: `4448`.
Counts are $(64 + 1) \cdot 16 = 1040$, $(3 \cdot 3 \cdot 16 + 1) \cdot 16 = 2320$, and $(16 + 1) \cdot 64 = 1088$. Total = 4448. The identity shortcut has no parameters.
Topic: architectures. Sources: HC5b pp.70–73; HC6a pp.107–111; S5.

Q48 — 275

Submit 275. Unrounded reference: `275`.
Each filter has $3 \cdot 3 \cdot 3 \cdot 2 = 54$ weights. Five filters and five biases give $(54 + 1) \cdot 5 = 275$. Do not count frames as independent input channels for this 3D convolution.
Topic: cnn. Sources: S5 Exercise 5.

Q49 — 0.36

Submit 0.36. Unrounded reference: `0.36`.
Box confidence = $.75 \cdot .6 = .45$. Class-specific score = $.8 \cdot .45 = .36$. The convention is explicitly given because different YOLO versions use different outputs and targets.
Topic: detection. Sources: HC6a pp.66–72; SOURCE_NOTES.

Q50 — 0.31

Submit 0.31. Unrounded reference: `0.3125`.
Sensitivity magnitude is $\frac{180}{144} = 1.25$ metres per pixel. Multiply by .25: .3125 metres, so submit .31. Depth becomes increasingly sensitive to disparity errors at small disparity/large depth.
Topic: stereo. Sources: HC6b pp.84–90; Szeliski §12.1; prerequisite calculus.